

Lecture 7 - Local Average Treatment Effects

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- Continue our discussion of causal effects under the potential outcomes framework $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$
- We have shown that the average treatment effect $E[\beta_i] = E[Y_i(1) - Y_i(0)]$ can be point identified in the following cases:
 - Randomized control trials, where $(Y_i(0), Y_i(1)) \perp D_i$
 - Selection on observables: $(Y_i(0), Y_i(1)) \perp D_i | X_i$ (can estimate CATE first or estimate propensity scores, and form $\frac{D_i Y_i}{\hat{p}_i}, \frac{(1-D_i) Y_i}{1-\hat{p}_i}$)
 - IV method if RCT has incomplete compliance: constant treatment effect $\beta_1 \equiv Y_i(1) - Y_i(0)$ can be identified via an instrument Z_i for D_i :

$$\hat{\beta}_1^{IV} = \frac{\bar{Y}^1 - \bar{Y}^0}{\bar{D}^1 - \bar{D}^0} \xrightarrow{p} \beta_1 \text{ constant TE} \sim \text{ATE}$$

where $\bar{Y}^1 = \frac{1}{N_1} \sum_{i: Z_i=1} y_i$ (sample mean among those randomly assigned to $Z_i = 1$).

- This lecture we continue to work with the general framework allowing for treatment effect heterogeneity:

$$\beta_{1i} := Y_i(1) - Y_i(0)$$

- Can we identify ATE $E[\beta_{1i}]$ using the IV method, or the ATE for some sub-population? What does $\hat{\beta}_1^{IV}$ converge to?
- Local average treatment effects (IV-LATE)
 - ATE for compliers (but not always takers/never takers/defiers)
 - characterize compliers, in comparison with other groups.
 - Application: Self-Sufficiency Project, C-section

Heterogeneous Treatment Effects

- Causal model of interest:

$$\begin{aligned}Y_i &= D_i Y_i(1) + (1 - D_i) Y_i(0) \\&= Y_i(0) + D_i \times (Y_i(1) - Y_i(0)) \\&= Y_i(0) + D_i \times \beta_{1i}\end{aligned}\tag{1}$$

- Define ATE $\beta_1 = E[\beta_{1i}]$, and $\beta_0 := E[Y_i(0)]$, (1) can be rewritten as:

$$\begin{aligned}Y_i &= \beta_0 + \beta_1 D_i + u_i \\ \text{where } u_i &= Y_i(0) - \beta_0 + (\beta_{1i} - \beta_1) D_i\end{aligned}$$

the residual u_i is clearly correlated with D_i unless $D_i \perp (Y_i(0), Y_i(1))$ (as in a perfect RCT).

- If you estimate a population regression $Y_i = b_0 + b_1 D_i + \epsilon_i$, what is b_1 ?

Instrument Variable Approach

- Suppose we have a valid instrument for treatment D_i : Z_i predicts D_i and $E[Z_i u_i] = 0$. Turns out the IV approach can help us get consistent estimates for the average treatment effects in some **sub-population (compliers)**.
- Let's look at an *experiment to study the effect of smoking by mothers on birthweight*.
 - Why not a RCT? Can't command mothers-to-be to smoke or not. So: conduct an “encouragement design” experiment among pregnant women who smoke and want to quit.
 - Z_i = **treatment** status = 1 if given “treatment” (counseling)
 - D_i = dummy for “desist” or stop smoking (1=desist, 0=keep smoking)

Instrument Variable Approach

- Y_i = weight of newly born infant

$$Y_i = \beta_0 + \beta_1 D_i + u_i \quad (\text{causal model}) \quad (2)$$

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (\text{1st stage})$$

$$Y_i = \underbrace{\beta_0 + \beta_1 \pi_0}_{\delta_0} + \underbrace{\beta_1 \pi_1}_{\delta_1} Z_i + \underbrace{(u_i + \beta_1 \eta_i)}_{\nu_i}$$

by exclusion restriction and the first-stage regression, $Z_i \perp (u_i, \eta_i)$.

Therefore, we can consistently estimate the reduced-form regression:

$$Y_i = \delta_0 + \delta_1 Z_i + \nu_i$$

$$\delta_1 = \beta_1 \pi_1$$

- This is the “dummy-dummy” case (both D_i and Z_i are binary).
What are first stage and reduced form coefficients in sample?

$$\begin{aligned} \hat{\pi}_1 &= \overline{D^1} - \overline{D^0} \\ \hat{\delta}_1 &= \overline{Y^1} - \overline{Y^0} \\ \Rightarrow \hat{\beta}_1^{iv} &= \frac{\overline{Y^1} - \overline{Y^0}}{\overline{D^1} - \overline{D^0}} \end{aligned} \quad (3)$$

Instrument Variable Approach

- We are dividing the difference in birthweights (between treatment and control groups) by the difference in the fraction of mothers who quit.
- What does $\hat{\beta}_1^{iv}$ represent? From (3), note $\hat{\beta}_1^{iv}$ is driven by the gains in birth outcomes for the **moms who quit smoking because of the encouragement**.
- Let's formalize this interpretation!

Instrument Variable in Potential Outcome Framework

- Let's think about how people behave and how they can respond to "treatment" ($Z_i = 1$, e.g. receive counseling).
- Let's introduce 2 new random variables in the potential outcome framework – that describe types of people and how they react: $(D_i(0), D_i(1))$
 - $D_i(0)$ = indicator of desist if person i is assigned to $z = 0$
 - $D_i(1)$ = indicator of desist if person i is assigned to $z = 1$
- Now we can characterize/classify individuals based on their **compliance type** $(D_i(0), D_i(1))$:
 - $D_i(0) = 0, D_i(1) = 0$ – "never takers" - never desist
 - $D_i(0) = 1, D_i(1) = 1$ – "always takers" - always desist
 - $D_i(0) = 0, D_i(1) = 1$ – "compliers" - desist if treated
 - $D_i(0) = 1, D_i(1) = 0$ – "defiers" - desist if untreated, not if treated.
- **(Random assignment)** All 4 groups should be equally likely in T and C groups.

$$(D_i(0), D_i(1)) \perp Z_i$$

which implies $\Rightarrow Pr(\text{Compliers}|Z_i) = Pr(\text{Compliers})$, and also for AT/NT/Defiers.

IV - First Stage

- Who is still smoking when kid is born?
 - in Control-group ($Z_i = 0$) : smokers = $NT's + Comp's$
 - in Treated-group ($Z_i = 1$) : smokers = $NT's$ (if no defiers)
- So: if no defiers, difference in fraction of smokers = fraction of **Compliers!** The first-stage under the random assignment assumption:

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$\pi_1 = E[D_i | Z_i = 1] - E[D_i | Z_i = 0]$$

$$= E[D_i(1) | Z_i = 1] - E[D_i(0) | Z_i = 0]$$

$$= Pr(D_i(1) = 1 | Z_i = 1) - Pr(D_i(0) = 1 | Z_i = 0)$$

$$= Pr(D_i(1) = 1) - Pr(D_i(0) = 1) \text{ by random assignment}$$

$$= Pr(AT + Comp) - Pr(AT + Defiers)$$

$(D_i(0), D_i(1))$
Independent of Z_i

The assumption $Pr(Defiers) = 0$ is called the **monotonicity assumption**:

$$\forall i : D_i(1) \geq D_i(0)$$

IV - First Stage

- note: if we flip the definition of $D_i = \text{smoking}$, this assumption is $\forall i : D_i(1) \leq D_i(0)$. There is no defier with $D_i(1) = 1$ while $D_i(0) = 0$.
- In summary, under the random assignment + monotonicity assumptions:

$$\pi_1 = Pr(Comp)$$

the first-stage coefficient is the fraction of compliers (characterized by $D_i(1) = 1, D_i(0) = 0$) in the population.

IV - 2nd Stage $Y_i(d, z)$

- Clearly the difference in birth weights between the $z = 0$ and $z = 1$ groups depends on the changes experienced by the compliers as a result of the intervention in the $z = 1$ environment. To formalize that we revise the *potential outcomes* for y :

$$Y_i(d, z) = \text{outcome for } i \text{ if } D_i = d \text{ and } Z_i = z$$

Now if z has no direct effect on the outcome:

$$\begin{aligned} Y_i(0, 0) &= Y_i(0, 1) = Y_i(0) \text{ the outcome if } D_i = 0 \\ Y_i(1, 0) &= Y_i(1, 1) = Y_i(1) \text{ the outcome if } D_i = 1 \end{aligned}$$

where $Y_i(d)$ is the potential outcome when unit i has $D_i = d$ (as before).

- We often write the **exclusion restriction** assumption as:

$$Y_i(d, z) = Y_i(d)$$

IV - 2nd Stage $Y_i(d, z)$

- Recap: Part of what it means to be a “good instrument” is that - z only works through its effect on D . We will assume this “exclusion restriction” is true. When we write

$$Y_i = \beta_0 + \beta_1 D_i + u_i \quad (\text{causal model})$$

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (\text{1st stage})$$

we are *building in the exclusion restriction*. Note that

- Z_i does not appear in the structural model
- since $E[Z_i u_i] = 0$, changes in z_i do not affect u_i . Thus the ONLY way for Z_i to affect Y_i is via the effect on X_i .

$$\left(\underset{\substack{| \\ d=0}}{K_i(0)}, \underset{\substack{| \\ d=1}}{K_i(1)}, \underset{\substack{| \\ z=0}}{D_i(0)}, \underset{\substack{| \\ z=1}}{D_i(1)} \right) \perp\!\!\!\perp Z_i$$

IV - 2nd Stage: Realized Y_i by group

The realized outcome for unit i can be expressed as:

$$Y_i = Y_i(D_i) = Y_i(D_i(Z_i)) \quad (4)$$


Let's break it down by group:

- Never takers ($D_i(0) = 0, D_i(1) = 0$): $D_i = 0$ regardless of $z_i \Rightarrow Y_i = Y_i(0)$ for either z_i .
- Compliers ($D_i(0) = 0, D_i(1) = 1$): observe $Y_i = Y_i(0)$ if $z_i = 0$, observe $Y_i = Y_i(1)$ if $z_i = 1$
- Always takers ($D_i(0) = 1, D_i(1) = 1$): $D_i = 1$ regardless of $z_i \Rightarrow Y_i = Y_i(1)$ for either z_i
- Defiers ($D_i(0) = 1, D_i(1) = 0$): observe $Y_i = Y_i(1)$ if $z_i = 0$, $D_i = 0$ if $z_i = 1 \Rightarrow$ observe $Y_i = Y_i(0)$ if $z_i = 1$

IV - 2nd Stage: Realized Y_i by group

$$Y_i = \delta_0 + \delta_1 Z_i + v_i, \quad \begin{cases} E[v_i] = 0 \\ E[v_i \cdot Z_i] = 0 \end{cases} \text{ orthog.} \Rightarrow \begin{cases} \delta_0 = E[Y_i | Z_i = 0] \\ \delta_1 = E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0] \end{cases}$$

Now let's think about the mean outcomes we observe in the $z = 0$ world:

$$\begin{aligned} E[Y_i | Z_i = 0] &= E[Y_i(D_i(0)) | Z_i = 0] \text{ by exclusion restriction } Y_i = Y_i(D_i) \\ &= E_g[E[Y_i(D_i(0)) | Z_i = 0, g]] \text{ by the law of iterated exp.} \\ &= \sum_{g \in \{NT, AT, Comp, Def\}} E[Y_i(D_i(0)) | g, Z_i = 0] \times Pr(g | Z_i = 0) \\ &= E[Y_i(0) | NT] \times Pr(NT) \\ &\quad + E[Y_i(0) | Comp] \times Pr(Comp) \\ &\quad + E[Y_i(1) | AT] \times Pr(AT) \\ &\quad + E[Y_i(1) | Def] \times Pr(Def) \end{aligned}$$

the last equation comes from the random assignment assumption $(D_i(1), D_i(0)) \perp Z_i$. So fractions of the various groups (e.g., $Pr(NT)$, etc) and expected potential outcomes (e.g., $E[Y_i(1) | AT]$, etc) do not depend on $z=0$ or 1 status.

IV - 2nd Stage: Realized Y_i by group

Likewise in the $z = 1$ world: *use the same steps before*

$$\begin{aligned} E[Y_i|Z_i = 1] &= E[Y_i(0)|NT] \times Pr(NT) \\ &\quad + E[Y_i(1)|Comp] \times Pr(Comp) \\ &\quad + E[Y_i(1)|AT] \times Pr(AT) \\ &\quad + E[Y_i(0)|Def] \times Pr(Def) \end{aligned}$$

If there are no defiers (under monotonicity assumption $D_i(1) \geq D_i(0)$):

$$\begin{aligned} E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] &= E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp) \\ &\quad + \underbrace{E[Y_i(0) - Y_i(1)|Defiers]}_{=0} \times Pr(Def) \end{aligned}$$

IV - 2nd Stage: Realized Y_i by group

Thus the population regression version of the reduced form model yields

$$\delta_1 = E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp)$$

and the population regression version of the 1st stage model yields

$$\pi_1 = E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = Pr(Comp)$$

So

$$\frac{\delta_1}{\pi_1} = E[Y_i(1) - Y_i(0) | Comp]$$

Under our assumptions, the ratio of the reduced form effect δ_1 to the 1st stage effect π_1 is the average treatment effect for compliers, which is called the local average treatment effects (local to compliers with $D_i(1) > D_i(0)$).

IV-LATE: Summary

Let's summarize the assumptions we have made in a IV-LATE framework, where we can classify individuals into four groups based on

$$(D_i(0), D_i(1)) = \begin{cases} (0, 0) & \text{never takers} \\ (0, 1) & \text{compliers} \\ (1, 0) & \text{defiers} \\ (1, 1) & \text{always takers} \end{cases}$$

- 1 **Random assignment:** treatment Z_i is randomly assigned. All four groups are equally likely to be in treated or in control.

$$(D_i(0), D_i(1)) \perp Z_i$$

- 2 **Exclusion Restriction:** (instrument) Z_i matters for outcome Y_i only through D_i

$$Y_i(d, z) = Y_i(d)$$

- Note 1 & 2 implies the joint distribution $(D_i(0), D_i(1), Y_i(0), Y_i(1))$ is the same whether $Z_i = 0$ or $Z_i = 1$.

- 3 **Monotonicity:** $D_i(1) \geq D_i(0)$ for all units. In other words, there are no defiers.

IV-LATE: Summary

Under the assumptions above, the first-stage regression gives us:

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$\pi_1 = E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = Pr(Comp)$$

and the reduced-form regression:

$$Y_i = \delta_0 + \delta_1 Z_i + \nu_i$$

$$\delta_1 = E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]$$

$$= E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp)$$

therefore the ratio, which is the population “IV”:

$$\beta_1^{IV} = \frac{\delta_1}{\pi_1} = E[Y_i(1) - Y_i(0)|Comp]$$

The IV estimate provides an estimate of the “gain” in Y that compliers obtain as a result of being in the $z = 1$ setting and getting $D_i = 1$ instead of $D_i = 0$.

- Take a closer look at the original model (heterogeneous treatment effects):

$$Y_i = \underbrace{\beta_{0i}}_{Y_i(0)} + \underbrace{\beta_{1i}}_{Y_i(1)-Y_i(0)} D_i = \beta_0 + \beta_1 D_i + u_i$$

in which

$$\beta_0 = E[Y_i(0)]$$

$$\text{ATE} \sim \beta_1 = E[Y_i(1) - Y_i(0)] = E[\beta_{1i}]$$

$$u_i = Y_i(0) - \beta_0 + D_i \times (\beta_{1i} - \beta_1)$$

D_i is correlated with u_i if $\beta_{1i} \neq \beta_1$ (heterog treatment effects) or $\text{cov}(D_i, Y_i(0)) \neq 0$ (selection into treatment by $Y_i(0)$).

- IV estimator:

$$\hat{\beta}_1^{IV} \xrightarrow{P} E[Y_i(1) - Y_i(0) | \text{Comp}] \text{ LATE}$$

which is different from ATE β_1 for the entire population.

- When does LATE = ATE?

- ④ (Boring case) Constant treatment effect: $Y_i(1) - Y_i(0) \equiv \beta_1$, we showed in the previous lecture that IV yields a consistent estimate of the constant TE.
- ② Homogeneous TE across sub-populations:

$$\begin{aligned} E[Y_i(1) - Y_i(0)|AT] &= E[Y_i(1) - Y_i(0)|NT] \\ &= E[Y_i(1) - Y_i(0)|Comp] \\ \Rightarrow E[Y_i(1) - Y_i(0)|Comp] &= E[Y_i(1) - Y_i(0)] \end{aligned}$$

- Is it plausible $E[Y_i(1) - Y_i(0)|Comp] = E[Y_i(1) - Y_i(0)|AT] = E[Y_i(1) - Y_i(0)|NT]$?
- From the data, we can estimate $E[Y_i(0)|NT]$ and $E[Y_i(1)|AT]$.
Note:

$$E[Y_i|D_i = 0, Z_i = 1] = E[Y_i(0)|NT, Z = 1] = E[Y_i(0)|NT]$$

$$E[Y_i|D_i = 1, Z_i = 0] = E[Y_i(1)|AT, Z = 0] = E[Y_i(1)|AT]$$

and also $E[Y_i(0)|Comp]$, $E[Y_i(1)|Comp]$:

$$\begin{aligned} E[Y_i|D_i = 0, Z_i = 0] = & \frac{Pr(NT)}{Pr(NT) + Pr(Comp)} \times E[Y_i(0)|NT] + \\ & + \frac{Pr(Comp)}{Pr(NT) + Pr(Comp)} \times E[Y_i(0)|Comp] \end{aligned}$$

Question: how do you estimate $Pr(NT)$ or $Pr(AT)$?

So notice when compliers are placed in control, they switch:

$$Z_i = 1$$

$$Z_i = 0$$

	$D_i(1)$	$Y_i(D_i(1))$	$D_i(0)$	$Y_i(D_i(0))$	$D_i(1) \geq D_i(0)$
AT	1	$Y_i(1)$	1	$Y_i(1)$	
NT	0	0	0	0	
Compliers	1	1	0	0	
Defiers	0	0	1	1	

→ assumed anyway w/
monotonicity assumption

- Comparing $E[Y_i(0)|NT]$ with $E[Y_i(0)|Comp]$, and $E[Y_i(1)|AT]$ with $E[Y_i(1)|Comp]$, if they are substantially different in levels, then it is much less plausible the ATE for compliers is similar to the average effects for other compliance types.

Example: Smoking

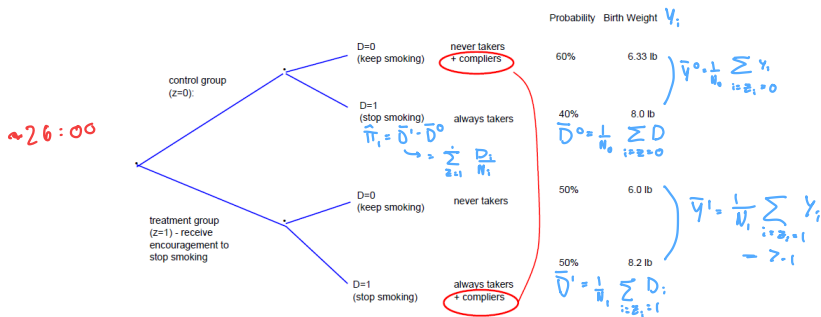
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Let's revisit the *experiment to study the effect of smoking by mothers on birthweight* (Y_i). $Z_i = 1$ if given "treatment" (counseling). D_i = dummy for "desist" or stop smoking (1=desist, 0=keep smoking)

- First stage: $\hat{\pi}_1 = \bar{D}^1 - \bar{D}^0 = 0.5 - 0.4 = \text{Pr}(\text{Comp})$

- Second stage:

$$\hat{\delta}_1 = \bar{Y}^1 - \bar{Y}^0 = (6 * 0.5 + 8.2 * 0.5) - (6.33 * 0.6 + 0.40 * 8) \approx 0.10$$



$$E[D | z=1] - E[D | z=0] = 0.1 \rightarrow \text{Pr}(\text{Complier}) = .1$$

$$E[Y | z=1] = 7.1 \text{ lb}; \quad E[Y | z=0] = 7.0 \text{ lb}$$

$$\beta_{IV} = (7.1 - 7.0) / (0.50 - 0.40) = 1.0$$

$$\beta_{IV} = \frac{\hat{\delta}_1}{\hat{\pi}_1} = (\bar{Y}^1 - \bar{Y}^0) / (\bar{D}^1 - \bar{D}^0)$$

1st stage reg: $D_i = \hat{\pi}_0 + \hat{\pi}_1 z_i + \epsilon_i$

$$\Rightarrow \hat{\pi}_0 = \bar{D}^0 = \frac{1}{N_0} \sum_{i: z_i=0} D_i$$

$$N_0 = \sum_{i: z_i=0} 1$$

(all the control units)

$$\Rightarrow \hat{\pi}_1 = \bar{D}^1 - \bar{D}^0$$

reduced form

$$y_i = \hat{\delta}_0 + \hat{\delta}_1 z_i + v_i$$

$$\hat{\delta}_0 = \bar{y}^0$$

$$\hat{\delta}_1 = \bar{y}^1 - \bar{y}^0$$

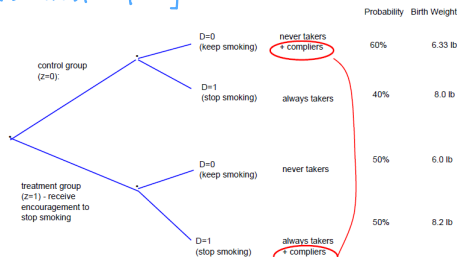
$$\text{ATE} = \frac{\hat{\delta}_1}{\hat{\pi}_0}$$

Example: Smoking

Is it plausible the ATE for AT or NT are similar to LATE?

- $E[Y_i(1)|AT]$ is approximated by the mean when $D = 0, Z = 1$: 8 lbs
- $E[Y_i|Z_i = 1, D_i = 1] = \frac{Pr(AT)}{Pr(AT)+Pr(Comp)} \times E[Y_i(1)|AT] + \frac{Pr(Comp)}{Pr(AT)+Pr(Comp)} \times E[Y_i(1)|Comp] \Rightarrow (8.2 - \frac{0.4}{0.5} * 8)/0.2 = 9$ is a consistent estimate of $E[Y_i(1)|Comp]$
- $E[Y_i(0)|Comp] = E[Y_i(1)|Comp] - \beta_1^{IV}$: about 8 lbs

$$\hat{\beta}_1^{IV} = \frac{\hat{\delta}_1}{\hat{\delta}_0} = \frac{\bar{y}^1 - \bar{y}^0}{\bar{D}^1 - \bar{D}^0} = E[Y_i(1) - Y_i(0) | \text{Compliers}]$$



$$E[D | z=1] - E[D | z=0] = 0.1 \rightarrow Pr(\text{Complier}) = 0.1$$

$$E[Y | z=1] = 7.1 \text{ lb}; E[Y | z=0] = 7.0 \text{ lb}$$

$$\beta_{IV} = (7.1 - 7.0) / (0.50 - 0.40) = 1.0$$

Characterize Compliers vs. AT, NT

- Let X_i denote some characteristics for unit i . Another way to see if make sense to extrapolate LATE to other compliance types is to compare observable characteristics of compliers vs. others. It's helpful to know how compliers as a subpopulation are compared with the overall population.
- We can estimate the following from the data:

$$E[X_i|D_i = 0, Z_i = 1] = E[X_i|NT]$$

apply LIE
Z_i instr. b/c random assignment

Ds at
coda

$$E[X_i|D_i = 0, Z_i = 0] = E[X_i|Comp] \times \frac{Pr(Comp)}{Pr(Comp) + Pr(NT)} + E[X_i|NT] \times \frac{Pr(NT)}{Pr(Comp) + Pr(NT)}$$

$$E[X_i|D_i = 1, Z_i = 0] = E[X_i|AT]$$

$$E[X_i|D_i = 1, Z_i = 1] = \dots$$

You could calculate the mean of X by (D, Z) and work out $\bar{X}^{Comp}$.

⊕ How do we get $E[X_i | D_i = 0, Z_i = 0]$?

• What does $E[X_i | D_i = 0, Z_i = 0]$ represent?

↳ A: X_i = some observed characteristic of mother

$D_i = 1$ if quits smoking (desists); 0 else

$Z_i = 1$ if she is encouraged to quit (recommended treatment)

$\Rightarrow E[X_i | D_i = 0, Z_i = 0]$ = "Among mothers in the control group ($Z=0$) who do not quit ($D_i=0$), what is the average of [characteristic X_i]"

What Prof wrote on board:

$$\begin{aligned} E[X_i | D_i = 0, Z_i = 0] &= E_g[E[X_i | D_i = 0, Z_i = 0, g]] \\ & \quad g_i(D_i(0), D_i(1)) \\ &= E_g[E[X | D(0) = 0, g]] \end{aligned}$$

$Z \perp\!\!\!\perp D_i(0), D_i(1)$

$Z \perp\!\!\!\perp X$

$$= \sum_g \Pr(g | D(0) = 0) \times E[X | g]$$

Characterize Compliers vs. AT, NT

- But there is an easier way! Define outcome $Y_i = X_i D_i$. Consider this *goofy* 2SLS model:

$$\text{Causal Model: } X_i D_i = a + b D_i + e_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (5)$$

which yields a reduced-form regression of $X_i D_i$ on instrument Z_i

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$$X_i D_i = \delta_0 + \delta_1 Z_i + \nu_i$$

$$\begin{aligned} \delta_1 &= E[X_i D_i | Z_i = 1] - E[X_i D_i | Z_i = 0] \quad \text{BY LIE} \\ &= E_g[E[X_i D_i(1) | Z_i = 1, g]] - E_g[E[X_i D_i(0) | Z_i = 0, g]] \end{aligned}$$

- Under random assignment, Z_i is independent of X_i . What does it imply?

Characterize Compliers vs. AT, NT

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- Also note $X_i D_i(1) = X_i$ if $D_i(1) = 1$, 0 otherwise, and $X_i D_i(0) = X_i$ for always takers.

where $g \in \{AT, NT, compliers\}$

$$E[X_i D_i | Z_i = 0] \sim E_g[E[X_i D_i(0) | Z_i = 0, g]] = E_g[E[X_i D_i(0) | g]] = \sum_g Pr(g) \cdot E[X_i D_i(0) | g] \\ = Pr(AT) \times E[X_i | AT]$$

$$E[X_i D_i | Z_i = 1] \sim E_g[E[X_i D_i(1) | Z_i = 1, g]] = E_g[E[X_i D_i(1) | g]] \\ \text{dropped by} \quad = Pr(AT) \times E[X_i | AT] \\ \text{random assignment} \quad + Pr(Comp) \times E[X_i | Comp]$$

Therefore:

$$\delta_1 = \frac{E[X_i D_i | Z_i = 1] - E[X_i D_i | Z_i = 0]}{Pr(Comp)} \\ b^{IV} = \frac{\delta_1}{\pi_1} = E[X | Comp] \quad \pi_1 \text{ (first stage)}$$

we have got the mean characteristics of compliers!

Application: $(Y_i(0), Y_i(1))$ as Characteristics



- Potential outcomes of each unit i can be thought of as characteristics X_i . We can apply/adjust the goofy model above to estimate $E[Y_i(0)|Comp]$, $E[Y_i(1)|Comp]$.
- Note $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0) \sim$ we observe $Y_i(1)$ among units with $D_i = 1$, and $Y_i(0)$ among $D_i = 0$.
- First, construct outcome

$$Y_i(1) \times D_i = \begin{cases} Y_i & D_i = 1 \\ 0 & D_i = 0 \end{cases}$$

and consider “goofy” 2SLS:

$$\text{Causal Model: } Y_i(1)D_i = a_1 + b_1 D_i + \xi_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (6)$$

Note: the proof is green in next slide

Application: $(Y_i(0), Y_i(1))$ as Characteristics

* By random assignment

Reduced-form $Y_i(1)D_i = \delta_0 + \delta_1 Z_i + \nu_i$ can be consistently estimated, in which

-27.00
Add 10 2

$$E[Y_i(1) \cdot D_i(1) | Z_i = 1]$$

$$E[Y_i(1) \cdot D_i(0) | Z_i = 0]$$

$$D_i = D_i(Z_i)$$

$$\begin{aligned}\delta_1 &= E[Y_i(1)D_i | Z_i = 1] - E[Y_i(1)D_i | Z_i = 0] \\ &= E[Y_i(1) | Comp] \times Pr(Comp)\end{aligned}$$

$$b_1^{IV} = E[Y_i(1) | Comp]$$

Application: $(Y_i(0), Y_i(1))$ as Characteristics

~~Start~~ here Mon 10/27

- Second, construct outcome

$$Y_i(0) \times (1 - D_i) = \begin{cases} 0 & D_i = 1 \\ Y_i(0) & D_i = 0 \end{cases}$$

$= Y_i$

and revise the goofy 2SLS:

$$\text{Causal Model: } Y_i(0)(1 - D_i) = a_0 + b_0 D_i + \xi_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (7)$$

the reduced-form coef. on Z_i becomes:

when $Z_i = 1, D_i = 0$: defiers

~~Office hours~~
wtf?

$$\begin{aligned} \delta_1 &= E[Y_i(0)(1 - D_i) | Z_i = 1] - E[\overbrace{Y_i(0)}^{y_i}(1 - D_i) | Z_i = 0] \\ &= -E[Y_i(0) | \text{Comp}] \times \text{Pr}(\text{Comp}) \\ b_0^{IV} &= -E[Y_i(0) | \text{Comp}] \end{aligned}$$

Application: $(Y_i(0), Y_i(1))$ as Characteristics

- The IV estimates b_1^{IV}, b_0^{IV} are linked to LATE:

$$E[Y_i(1) - Y_i(0)|Comp] = b_1^{IV} + b_0^{IV}$$

Is this surprising?

$$Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$$

we estimate the same first stage, 3 reduced form on **the same sample**, where the dependent variables $Y_i(1)D_i + Y_i(0)(1 - D_i)$ add up to Y_i :

$$Y_i = \delta_0 + \delta_1 z_i + \nu_i$$

$$Y_i(1)D_i = \delta_0^{(1)} + \delta_1^{(1)} z_i + \nu_i^{(1)}$$

$$Y_i(0)(1 - D_i) = \delta_0^{(0)} + \delta_1^{(0)} z_i + \nu_i^{(0)}$$

$$Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$$

$$= \left(\delta_0^{(1)} + \delta_0^{(0)}\right) + \left(\delta_1^{(1)} + \delta_1^{(0)}\right) z_i + \left(\nu_i^{(1)} + \nu_i^{(0)}\right)$$

Application: $(Y_i(0), Y_i(1))$ as Characteristics

with $(\delta_1^{(1)} + \delta_1^{(0)}) = \delta_1 = E[Y_i(1) - Y_i(0) | Comp] \times Pr(Comp)$ as the instrument $z_i \perp (\nu_i^{(1)} + \nu_i^{(0)})$ (orthogonal).

Application: Self-Sufficiency Program

~31:00

Audio 2

Example from Card and Hyslop (2005):

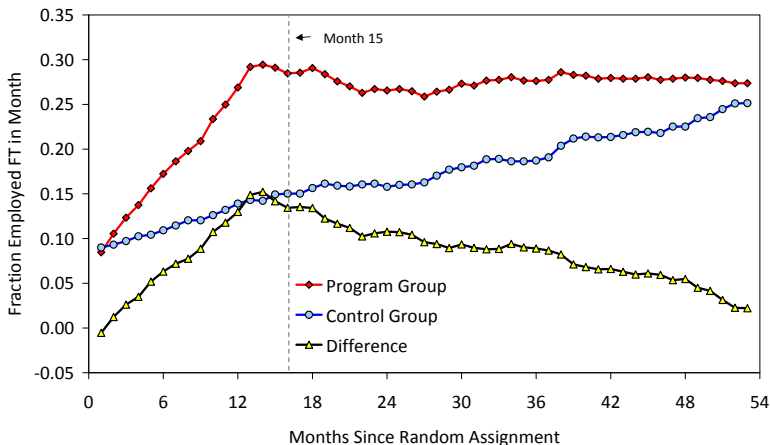
- Self Sufficiency Project (SSP) experiment, conducted in Canada, early 1990s
- Earnings subsidy (negative tax) for people who enter full-time (FT) work
- Subsidy = $0.5 \times (T - \text{monthly earnings})$ $T = \$2500/\text{mo}$ or $\$3083/\text{mo}$
- ONLY for the FT (30+ hrs/week) who earn $\geq$ the minimum wage:
 - subsidy lasts for 3 years;
 - have to start receiving subsidy within 12 mo. of random assignment.
- **Research Questions:** SSP subsidy = strong incentive to enter FT work, which is a “short-term establishment incentive”.
 - 1 can long-term welfare recipients find a job and leave welfare? (short-term establishment)
 - 2 can they maintain a FT job and be better off in the long-term? (long-term entitlement as they choose work over welfare)

Experiment:

- Welfare recipients with kids (95% females), 1+ years on welfare
- Randomly allocated to treatment group ($z = 1$) or control ($z = 0$)
- Series of surveys measure employment etc.

Application: Self-Sufficiency Program

Monthly FT Empl. Rates of SSP Control and Treatment Groups



Causal Model

- Outcome "Y": $IA15_i = 1$ if on welfare in month 15
- Endogenous variable of interest "D": $FT15_i = 1$ if the worker is employed full-time in month 15.
- Instrument "Z": random assignment to treatment group in the SSP experiment
- Causal model for receiving income assistance (IA) at month 15:

$$IA15_i = \beta_0 + \beta_1 FT15_i + u_i$$

- First stage model for FT empl at month 15:

$$FT15_i = \pi_0 + \pi_1 TREAT_i + \eta_i$$

- Reduced form model for IA at month 15:

$$IA15_i = \delta_0 + \delta_1 TREAT_i + \nu_i$$

- Underlying assumptions:

- ① Treatment is randomly assigned

$$(D_i(0), D_i(1)) \perp\!\!\!\perp Z_i \\ (FT(0), FT(1)) \perp\!\!\!\perp TREAT_i$$

- ② Exclusion restriction $Y(d, z) = Y(d)$: SSP only affects *IA* through full-time work? *SSP subsidy* $\Rightarrow$ *FT work* $\Rightarrow$ *leave IA* seems plausible.
- ③ Is “no defiers” satisfied? Likely since SSP would never cause you to stop working FT (unless you are really weird).

$$FT_{12i}(0) = 0, \quad FT_{12i}(1) = 1$$

↙ find diff horizon: can become never holder

$$t = 24: \quad FT_{24i}(0) = 0 = FT_{24i}(1)$$

$$E[D|Z=0] = E[D(0)|Z=0] \\ = \Pr(AT)$$

* make sure to use
LIE in the exam
to show this

★ Review the first-stage ↻

Estimate LATE

AT = always takers
C = compliers
NT = never takers

Summary of SSP Outcomes at Month 15

	Employed FT (1)	On Welfare (IA) γ (2)
$z=0 \sim$ Control Group (n=2718)	0.1472 $E[D z=0] = Pr(AT)$	0.8098 = $\bar{y}^0 \rightarrow E[y z=0]$
$z=1 \sim$ Treatment Group (n=2762)	0.2848 $E[D z=1] = Pr(AT) + Pr(C)$	0.6687 = $\bar{y}^1 \rightarrow E[y z=1]$
Difference: T-C (std error)	$\hat{\pi}_1$ 0.1376 (0.011)	IV Estimate -0.1411 (0.012)
IV estimate:	-1.027 (0.082)	$\bar{y}^1 - \bar{y}^0 = \hat{\delta}_1$ $y_i = \delta_0 + \hat{\delta}_1 z_i + v_i$

So NT: $1 - E[D|z=1] = 1 - 0.2848$ (b/c no defiers)

why can't we tell

- We know that there is only partial compliance with the subsidy incentive.

$$Pr(NT, mo.15) = 1 - 0.2848 = 0.7152$$

$$Pr(AT, mo.15) = 0.1472$$

$$Pr(C, mo.15) = 0.1376$$

- Note that these fractions are changing as the experiment goes on (x-axis = months since random assignment)

Q: who are the compliers? Are the “like” other people? Are they more or less advantaged?

- X_i = HS graduation
- Apply goofy 2SLS to outcome “D*X” $\sim X_i \times FT15_i$

$$X_i FT15_i = a + b \times FT15_i + u_i$$

$$FT15_i = \pi_0 + \pi_1 Z_i + \nu_i$$

$$\begin{aligned}
 b^{IV} &= \frac{\delta_1}{\pi_1} \text{ via reduced-form} \\
 &= \frac{E[X_i FT15_i | Z_i = 1] - E[X_i FT15_i | Z_i = 0]}{E[FT15_i | Z_i = 1] - E[FT15_i | Z_i = 0]} \\
 &= \frac{\text{cov}(X_i FT15_i, \widehat{FT15_i})}{\text{var}(\widehat{FT15_i})} \text{ via 2SLS}
 \end{aligned}$$

where predicted $\widehat{FT15_i} = \pi_0 + \pi_1 Z_i$. Note
 $\text{cov}(X_i FT15_i, \widehat{FT15_i}) = \pi_1 \times \text{cov}(X_i FT15_i, Z_i)$ and
 $\text{var}(\widehat{FT15_i}) = (\pi_1)^2 \times \text{var}(Z_i)$.

Audito 4
(if stopped in)

$$E[XD | Z=1] = E_g[E[X \cdot D(1) | Z=1, g]]$$

based on random assignment assumption:

$$(D_i(0), D_i(1)) \perp\!\!\!\perp Z_i$$

$$\text{if } D_i(D_i(Z))$$

you can it write

$$D_i \perp\!\!\!\perp Z_i$$

then,

$$E[XD | Z=1] = E_g[E[XD(1) | g]]$$

$$= Pr(AT) \cdot E[X \cdot 1 | AT] + Pr(C) \cdot E[X \cdot 1 | C]$$

$$\text{so for } E[XD | Z=0] = E_g[E[X \cdot D(0) | Z=0, g]]$$

$$= E_g[E[X \cdot D(0) | g]]$$

$$= Pr(AT) \cdot E[X \cdot 1 | AT]$$

by random assignment
 $(D_i(0), D_i(1)) \perp\!\!\!\perp Z_i$

so its $E[XD]$

Characterize Compilers

[9:45 AM]

EXAM Q !!!

Two-Stage Least Squares Estimation

Model d_hsggrad
Dependent Variable d_hsggrad

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	7.401383	7.401383	139.92	<.0001
Error	5478	289.7814	0.052899		
Corrected Total	5479	609.8392			

Root MSE	0.23000	R-Square	0.02491
Dependent Mean	0.12755	Adj R-Sq	0.02473
Coeff Var	180.31324		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variable Label
Intercept	1	0.011784	0.010269	1.15	0.2512	Intercept
EMFTM15	1	0.534926	0.045223	11.83	<.0001	Msurv:Employed full-time in mo15

• % of hs among compilers

⇒ AT: look at control group

Application: The Health Impacts of C-section

$$\frac{1}{2} \quad \frac{3}{4} \quad \frac{7}{8} \quad \frac{15}{16}$$

- Card, Fenizia, Silver: “The Health Impacts of Hospital Delivery Practices”
- Research Question: treatment practices vary widely across hospitals. In the context of childbirth, the authors assess impacts of the differences in delivery practices at childbirth on infant and maternal health
- Data: births in CA, with detailed information such as where patients live, where they deliver kids, delivery practices, health outcomes through records of ER visits and medical claims...
 - they classify hospitals based on c-section rates: H if a “high c-section” hospital, L otherwise
 - D_i = deliver at H hospital
 - y_i = c-section delivery; baby is readmitted to hospital after birth
- Empirical challenge: there is endogenous sorting in c-section (e.g., high risk, unobserved preferences...)

Application: The Health Impacts of C-section

Not on exam

- Quasi-experiment Design:
 - living closer to a H hospital than a L *exogenously* increases the chance that they choose a H hospital;
 - Z_i = relative distance of mom's home to H vs L hospital (continuous or indicator)
 - There are 2 types of compliers:
 - H -compliers: mothers who choose a H hospital because they live closer to H ;
 - $H\&C$ -compliers: mothers who choose a H hospital because they live closer to H and will choose c-section; if they live closer to a L they would not have used c-section.

First-stage: “H” or “C&H” Compliers

Assume: (1) random assign (2) exclusion restriction and (3) monotonicity.

deliver at H hospital: $H_i = \pi_0 + \pi_1 Z_i + \pi_X X_i + \epsilon_i$

deliver by C-section: $C_i = \gamma_0 + \gamma_1 Z_i + \gamma_X X_i + \eta_i$

↪ increase in c-section rate [7:50 Aud Q]

Interpretations:

- π_1 : the fraction of H-compliers, who will choose a H hospital if they live closer to H. Note this includes people who would have chosen C-section at both (H, L) hospitals (“C” always takers), and people who were to choose c-section if they chose a H-hospital (C&H compliers).
- γ_1 : the fraction of “C&H” compliers. Why? Under monotonicity assumption, the only group who changes C_i are C&H compliers who choose a H hospital because they live closer to H, and who choose C-section because they are at H-hospitals.

First-stage: “H” or “C&H” Compliers

Table III: Estimated Effects of Relative Distance on Place, Mode, and Timing of Delivery

Outcome Variable	Mean	Instrument=Relative Distance to H Hospital Coefficients × 100		Instrument= Indicator for Closer to H Hospital Coefficients × 10	
		Baseline controls only	All controls	Baseline controls only	All controls
	(1)	(2)	(3)	(4)	(5)
Deliver at H Hospital	0.515	1.304 (0.135)	1.328 (0.134)	0.682 (0.094)	0.694 (0.091)
C-section Delivery	0.256	0.159 (0.028)	0.149 (0.027)	0.078 (0.021)	0.080 (0.020)
<i>Breakdown of C-Section Deliveries:</i>					
C-Section at H Hospital	0.149	0.401 (0.041)	0.403 (0.041)	0.208 (0.030)	0.213 (0.030)
C-Section at L Hospital	0.106	-0.243 (0.033)	-0.254 (0.030)	-0.130 (0.024)	-0.133 (0.022)
<i>Fractions of Complier Groups -- Moving 7 mi. closer to H hospital (col. 2-3) or closer to H hospital (col. 4-5)</i>					
P(H Complier)		0.091	0.093	0.068	0.069
P(C&H Complier)		0.011	0.010	0.008	0.008
P(H Complier & C Always-Taker)		0.017	0.018	0.013	0.013

Defiers: choose C-section when go to low C-sect hospital

IV-LATE for “H” Compliers

1st stage: whether you deliver at a high-C-sect hospital w/ . . .

- Focusing on the H-hospital compliers, estimate the 2SLS model:

$$\begin{aligned}Y_i &= \beta_0 + \beta_1 H_i + \beta_X X_i + u_i \\&= \delta_0 + \delta_1 Z_i + \delta_X X_i + \nu_i \\ \Rightarrow \beta_1^{IV} &= \frac{\delta_1}{\pi_1}\end{aligned}$$

the IV estimate represents the LATE for H-compliers – the change in outcome y per additional delivery due to H hospital among the compliers.

- The authors find mixed health impacts: in the short run readmission rate is lower, but in the longer run ER visits are higher among the compliers.

IV-LATE for "H" Compliers

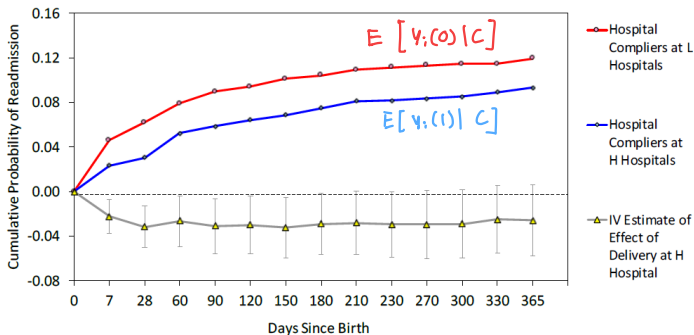
25:00

$E[Y_i(1)|C] \rightarrow$ to compute this, use the goofy 2SLS

$$\Rightarrow E[Y_i(1)|C] = \frac{E[Y_i D_i | Z=1] - E[Y_i D_i | Z=0]}{E[D_i | Z=1] - E[D_i | Z=0]}$$

go back to slide 31
app. $(Y_i(0), Y_i(1))$

a. Hospital Readmissions



Reduced form

$$\begin{aligned}\delta_1 &= E[Y_i(1)D_i | Z=1] - E[Y_i(1)D_i | Z=0] \\ &= E[Y_i(1) | \text{comp}] \times D_0(\text{comp})\end{aligned}$$

$$E[Y_i D_i | Z=0] = E_g[E[Y_i D_i(0) | Z=0, g]]$$

35:00

$$= E_g[\Pr(D_i(0)=1 | g, Z=0) \cdot E[Y_i | D_i(0)=1, Z=0, g]]$$

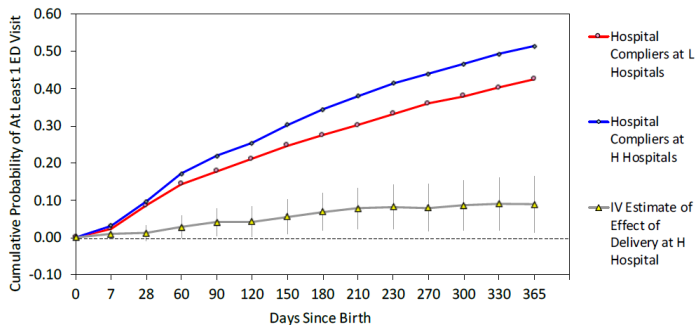
$\swarrow Y_i(D_i) = Y_i(D_i(0))$

b/c $E[Y_i(1) \cdot D_i(1) | Z=1], E[Y_i(1) \cdot D_i(0) | Z=1]$; $D_i := D_i(Z_i)$
under potential outcomes framework

$$= E_g[\Pr(D_i(0)=1 | g) \cdot E[Y_i(1) | D_i(0)=1, g]]$$

IV-LATE for “H” Compliers

b. Emergency Department Visits



- Question: How did they estimate $E[Y_i(1)|Comp]$, $E[Y_i(0)|Comp]$?
Hint: goofy 2sls on $Y_i D_i$ and $Y_i(1 - D_i)$

- The “relative distance” instrument is continuous: Z_i = distance to the closest Low-csection hospital relative to the closest H-hospital.
- But we can apply the goofy 2SLS to characterize the compliers:

$$\text{Causal Model: } X_i D_i = a + b D_i + e_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$b^{IV} = \frac{\text{cov}(X_i D_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{E[X(Z_i - E[Z_i]) | D = 1]}{E[Z_i - E[Z_i] | D = 1]}$$

Table IV: Characteristics of Compliers

	Analysis Sample	Compliers		Ratio: C's/All (%)
<u>Socio-Economic Characteristics of Mother</u>				
<u>Race/Ethnicity</u>				
White	31.7%	25.9	(5.1)	81.6
Black	5.6%	0.7	(1.5)	12.4
Asian	17.6%	11.3	(4.8)	64.1
Hispanic	44.2%	62.2	(5.7)	140.7
<u>Education</u>				
High School or Less	41.2%	54.4	(4.7)	132.0
Some College	20.1%	12.9	(2.1)	64.5
BA or Higher	38.7%	32.6	(5.3)	84.4
Government Insurance	43.3%	62.4	(5.7)	144.2
<u>Predicted Probabilities of Infant ED Visit/Inpatient Admissions</u>				
Prob(ED Visit) > median	50.0%	67.8	(6.3)	135.5
Prob(Inpat. Adm) > median	50.0%	71.0	(6.4)	142.1

46:00// Lec 8 →

$$\begin{aligned} b^{IV} &= \frac{\text{cov}(X_i D_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{E[X(Z_i - E[Z_i]) | D = 1]}{E[Z_i - E[Z_i] | D = 1]} \\ &= \int_{\underline{z}} E[X | \text{Comp}(\underline{z})] \times \omega(\underline{z}) d\underline{z} \end{aligned}$$

is weighted average of $E[X | \text{Comp}(\underline{z})]$ across complier groups, with weights $\int \omega(\underline{z}) d\underline{z} = 1$.

- Think of $\text{Comp}(\underline{z})$ as compliers who would have $D_i(\underline{z}) = 1[Z_i \geq \underline{z}]$ – folks who would deliver at H-hospitals if the closest L-hospital is $\underline{z}$ miles further than the closest H.
- For always takers, $\underline{z} = -\infty$, and for never takers, $\underline{z} = \infty$.
- In the binary instrument case, there is only 1 group of compliers: $D_i(1) > D_i(0)$.

Characterize Compliers - Continuous Instrument

- Let's classify units by $\underline{z}$ (resistance to treatment) where $D_i(\underline{z}) = 1[Z_i \geq \underline{z}]$
- Denominator (applying random assign of Z): let $\tilde{Z}_i = Z_i - E[Z_i]$, and F represent the cdf of the $\underline{z}$,

$$\begin{aligned} E[\tilde{Z}|D = 1] &= E_{\underline{z}}[E[\tilde{Z}|D = 1, \underline{z}]] \text{ by the law of iterated exp.} \\ &= \int_{\underline{z}} E[\tilde{Z} | \underbrace{Z \geq \underline{z}}_{D=1}, \underline{z}] dF(\underline{z}) = \int_{\underline{z}} E[\tilde{Z} | Z \geq \underline{z}] dF(\underline{z}) \end{aligned}$$

- Numerator:

$$\begin{aligned} E[X\tilde{Z}|D = 1] &= \int_{\underline{z}} E[X\tilde{Z}|Z \geq \underline{z}, \underline{z}] dF(\underline{z}) \\ &= \int_{\underline{z}} E[X|\underline{z}] \times E[\tilde{Z}|Z \geq \underline{z}] dF(\underline{z}) \end{aligned}$$

by random assign X independent of Z

Characterize Compliers - Continuous Instrument

- So the ratio

$$\frac{E[X\tilde{Z}|D=1]}{E[\tilde{Z}|D=1]} = \int_{\underline{z}} E[X|\underline{z}] \times \underbrace{\frac{E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})}{\int_{\underline{z}} E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})d\underline{z}}}_{\text{weight } \omega(\underline{z})} d\underline{z}$$

where the weights sum to 1:

$$\int_{\underline{z}} \omega(\underline{z})d\underline{z} = \frac{\int_{\underline{z}} E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})d\underline{z}}{\int_{\underline{z}} E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})d\underline{z}} = 1.$$